

RM-201 Indoor Air Quality Control - Owner Requirements

Document URS-IAQ-001 - Revision A - 2026-01-08

1. Scope

This specification defines the owner requirements for the RM-201 single-zone demand-controlled outdoor-air demonstration. The system shall represent a room, an outdoor-air source, an exhaust boundary, occupant CO₂ generation, a room CO₂ measurement, and a feedback controller that adjusts ventilation.

2. Functional requirements

URS-IAQ-001 - The room shall be represented as a single well-mixed air volume of 100 m³.

URS-IAQ-002 - Outdoor air shall be supplied through a controlled airflow path and room air shall leave through an exhaust path.

URS-IAQ-003 - The ventilation controller shall modulate fresh-air flow using measured room CO₂ as feedback.

URS-IAQ-004 - During the approved occupied schedule, room CO₂ shall not exceed 1000 ppm.

URS-IAQ-005 - The reference outdoor-air CO₂ concentration shall be 300 ppm.

URS-IAQ-006 - CO₂ generation shall be proportional to the scheduled number of occupants.

URS-IAQ-007 - The design occupant CO₂ generation rate shall be 8.18E-6 kg/s per person.

URS-IAQ-008 - The released demonstration shall support a peak occupancy of at least 12 persons. Final operations schedule to be issued separately.

3. Control and verification expectations

URS-IAQ-009 - A numerical simulation shall demonstrate the CO₂ feedback behavior over a representative daily occupancy schedule.

URS-IAQ-010 - Concentration and airflow quantities shall be represented with explicit engineering units or unambiguous normalization definitions.

URS-IAQ-011 - The model shall distinguish physical outdoor-air flow direction from any implementation-specific source sign convention.

URS-IAQ-012 - Verification evidence shall record occupant count, room CO₂, outdoor CO₂, commanded ventilation, and physical ventilation flow.

Revision note. Rev A is the baseline owner document. Detailed controls tuning, the final occupancy schedule, and resolution of ambiguous wording may be established by later approved engineering records.